Linear Algebra: Guided Review and Exercises

Purpose. This sheet is meant to be a learning document. Some of the computational procedures below were not developed in detail in lecture. Each such topic therefore begins with a short introduction and a worked example, followed by exercises.

Unless otherwise stated, all vector spaces are finite-dimensional over $\mathbb{R}$ or $\mathbb{C}$.

I. Solving Linear Systems by Elimination

Mini-lecture. A linear system is best viewed through its *augmented matrix*. The three elementary row operations are:

(i) interchange two rows;
(ii) multiply a row by a nonzero scalar;
(iii) add a multiple of one row to another row.

These operations do not change the solution set. The practical goal is echelon form: identify pivot variables, free variables, and any contradictory row of the form

$$(0 \ \cdots \ 0 \mid c), \qquad c \neq 0.$$

A contradictory row means no solution. If every variable is a pivot variable, the solution is unique. If the system is consistent and at least one variable is free, there are infinitely many solutions.

Worked example. Consider

$$\begin{aligned} x + y + z &= 6, \\ 2x + y + z &= 8, \\ x + 2y + z &= 9. \end{aligned}$$

Start from

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & 1 & 8 \\ 1 & 2 & 1 & 9 \end{array}\right).$$

Use $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & -4 \\ 0 & 1 & 0 & 3 \end{array}\right).$$

Swap the last two rows and then add the new second row to the third:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & -1 \end{array}\right).$$

Thus $z = 1$, $y = 3$, and $x = 2$. The point of the procedure is not the arithmetic itself: the pivot structure tells us whether a solution exists and how many degrees of freedom it has.

1. **Practice.** For each system below:

(a) write the augmented matrix;

(b) row-reduce to echelon form;

(c) identify pivot and free variables;

(d) state whether the system has a unique solution, no solution, or infinitely many solutions;

(e) if there are infinitely many solutions, parameterize the complete solution set.

(i)

$$\begin{aligned} 2x + 2y &= 5, \\ x - 4y &= 0; \end{aligned}$$

(ii)

$$\begin{aligned} x - 3y + z &= 1, \\ x + y + 2z &= 14; \end{aligned}$$

(iii)

$$\begin{aligned} -x - y &= 1, \\ -3x - 3y &= 4. \end{aligned}$$

For every consistent nonunique system, write the solution in the form

$$x = x_p + x_h,$$

where x_p is one particular solution and x_h ranges over the null space of the coefficient matrix.

i. a)
b) $\left[\begin{array}{cc|c} 2 & 2 & 5 \\ 1 & -4 & 0 \end{array}\right] \xrightarrow{-\frac{1}{2}R_1} \left[\begin{array}{cc|c} 2 & 2 & 5 \\ 0 & -5 & -2\frac{1}{2} \end{array}\right]$

c) x, y pivots

d) unique, $y = 1/2$ $x = 2$

$\Rightarrow \vec{x} = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$

e) N/A

$x - 3y + z = 1,$
$x + y + 2z = 14;$

i. a, b)
$$\left[\begin{array}{ccc|c} 1 & -3 & 1 & 1 \\ 1 & 1 & 2 & 14 \end{array}\right] \xrightarrow{-R_1} \left[\begin{array}{ccc|c} 1 & -3 & 1 & 1 \\ 0 & 4 & 1 & 13 \end{array}\right]$$

c) x, y pivot, z free

d) infinite solutions

e) let $z = t$

$\Rightarrow y = 3t \Rightarrow \vec{x} = \begin{bmatrix} 4t \\ 3t \\ t \end{bmatrix}$

$x = 4t$

for $t = 1$, $\vec{x}_1 = \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$

All sol: $t \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$ for $t \in \mathbb{R}$

$-x - y = 1,$
$-3x - 3y = 4.$

ii. a, b)
$$\left[\begin{array}{cc|c} -1 & -1 & 1 \\ -3 & -3 & 4 \end{array}\right] \xrightarrow{-3R_1} \left[\begin{array}{cc|c} -1 & -1 & 1 \\ 0 & 0 & 1 \end{array}\right]$$

c) 1 pivot, 1 free

d) no solution, inconsistent

row $0x + 0y = 1$

e) N/A

2. A parameter and consistency. Consider

$$3x + 2y = 10,$$
$$6x + 4y = b.$$

$A = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$

(a) Row-reduce the augmented matrix without assigning a value to b at the outset.

(b) For which values of b is the system consistent?

(c) When it is consistent, parameterize all solutions.

(d) Compare the rank of the coefficient matrix with the rank of the augmented matrix in the consistent and inconsistent cases.

a) $A^* = \left[\begin{array}{cc|c} 3 & 2 & 10 \\ 6 & 4 & b \end{array}\right] \xrightarrow{-2R_1} \left[\begin{array}{cc|c} 3 & 2 & 10 \\ 0 & 0 & b-20 \end{array}\right] = R$

b) $b = 20$ only

c) Let $b = 20 \Rightarrow \left[\begin{array}{cc|c} 3 & 2 & 10 \\ 0 & 0 & 0 \end{array}\right]$

$3x + 2y = 10$

Let $y = t$ be free variable,

$x = \frac{10 - 2t}{3} = \frac{10}{3} - \frac{2}{3}t$

$\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{10}{3} - \frac{2}{3}t \\ 0 + 1t \end{bmatrix} = \begin{bmatrix} 10/3 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2/3 \\ 1 \end{bmatrix}$

$\vec{x} = \vec{x}_0 + t\vec{v}$

Consistent $b = 20$ inconsistent $b \neq 20$

d) $\rho(A) = 1$, $\rho(A^*) = 1$; $\rho(A) = 1$, $\rho(A^*) = 2$

II. Rank, Column Space, and Null Space

Mini-lecture. Row reduction reveals rank and the null space. If the reduced matrix has r pivot columns, then the matrix has rank r. To find a basis of the *column space*, use the corresponding pivot columns of the *original* matrix. To find the null space, solve $Ax = 0$ using the reduced system and parameterize by the free variables.

Worked example. Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}.$$

Row reduction gives

$$A \sim \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence rank $A = 2$. The first two columns are pivot columns, so a basis for the column space is formed by the first two columns of the *original* matrix:

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} \right\}.$$

For $Ax = 0$, write $x_3 = t$. Then $x_1 = t$ and $x_2 = -2t$, so

$$\operatorname{Ker} A = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right\}.$$

Notice the rank-nullity check: $2 + 1 = 3$, the number of columns of A.

3. Practice. For each matrix below, use row reduction to find:

(a) its rank;

(b) a basis for its column space;

(c) a basis for its null space;

(d) the dimensions predicted by rank-nullity.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 & -3 \\ -3 & 1 & 2 \\ 2 & -3 & 1 \end{pmatrix}.$$

Finally, explain how your calculations determine whether the columns of each matrix are linearly independent.

A.a) $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \xrightarrow{-3R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -5 & -7 \\ 2 & 3 & 1 \end{pmatrix} \xrightarrow{-2R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -5 & -7 \\ 0 & -1 & -5 \end{pmatrix}$

$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & -5 & -7 \end{pmatrix} \xrightarrow{-5R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & 0 & 18 \end{pmatrix}$

b) all pivots, all original cols.

$A_B = \left\{ \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right\}$

c) None (null space = 0)

d) dim = 3

B. a) $\begin{pmatrix} 1 & 2 & -3 \\ -3 & 1 & 2 \\ 2 & -3 & 1 \end{pmatrix} \xrightarrow{+3R_1} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 7 & -7 \\ 2 & -3 & 1 \end{pmatrix}$

$\xrightarrow{-2R_1} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 7 & -7 \\ 0 & -7 & 7 \end{pmatrix} \xrightarrow{+R_2,\ R_2/7} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$

b) 2 pivots, col 1 + 2

$B_B = \left\{ \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} \right\}$

c) $B\vec{x} = \vec{0}$ $\quad$ $x + 2y - 3z = 0$

$-3x + y + 2z = 0$

$2x - 3y + z = 0$

let $z = t$ $\quad$ $x = t,\ y = t$

$\operatorname{Ker} B = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$

d) $\dim B = 3 = \dim \operatorname{Col} B + \dim \operatorname{Ker} B$

4. Proof: rank-nullity without coordinates. Suppose $T : \mathbb{F}^4 \to \mathbb{F}^2$ is linear and

$$\operatorname{Ker} T = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_1 = 5x_2,\ x_3 = 7x_4\}.$$

Without constructing a matrix for T, prove that T is surjective.

Claim: T is surjective iff $\operatorname{Im} T = \mathbb{F}^2$, the entire codomain

By rank nullity thm,

$\dim \mathbb{F}^4 = \dim \operatorname{Im} T + \dim \operatorname{Ker} T$

Since $\operatorname{Ker} T = \{(5x_2, x_2, 7x_4, x_4)\}$, T maps all x_2 & x_4 to the zero vector in $\mathbb{F}^2$, s.t. $\dim \operatorname{Ker} T = 2$

$\dim \mathbb{F}^4 = 4 = \dim \operatorname{Im} T + 2$

s.t. $\dim \operatorname{Im} T = 2$

since $\dim \mathbb{F}^2 = 2$, $\operatorname{Im} T$ is the entire codomain, thus surj. $\square$

5. Proof: spanning and independence as properties of a map. Let $v_1, \ldots, v_m \in V$, and define

$$T : \mathbb{F}^m \to V, \qquad T(z_1, \ldots, z_m) = z_1 v_1 + \cdots + z_m v_m.$$

Prove directly from the definitions that

(a) $v_1, \ldots, v_m$ spans V if and only if T is surjective;

(b) $v_1, \ldots, v_m$ is linearly independent if and only if T is injective.

a) $\Rightarrow$ if $\text{span}(v_1, \ldots, v_m) = V$, then

$\forall v \in V$, $\forall z_i \in \mathbb{F}$, $v = z_1 v_1 + \ldots + z_m v_m$

since $T(z_1, \ldots, z_m) = \sum_{i=1} z_i v_i$,

$\text{span}(v_i) = V = z_1 v_1 + \ldots + z_m v_m = T(z_i)$

T surj. iff $\text{Im}\, T = V$

$\text{Im}\, T = T(z_1, \ldots, z_m) = V$

$\Rightarrow T$ surj.

$\Leftarrow$ if T surj, $\text{Im}\, T = V$

s.t. $T(z_1, \ldots, z_m) = V$

$z_1 v_1 + \ldots + z_m v_m = v$ for all $v \in V$

$V = \text{im}\, T = T(z_1, \ldots, z_m) = \text{span}(v_1, \ldots, v_m)$

thus $V = \text{span}(v_1, \ldots, v_m)$

b) ($\Rightarrow$) if $v_1, \dots, v_m$ lin ind.,
the only $z_i \in \mathbb{F}$ that solve
$z_1 v_1 + \dots + z_m v_m = 0$ are $z_1 = \dots = z_m = 0$
therefore the only lin comb $\sum_{i=1}^{m} z_i v_i = 0$
is $\vec{z} = (0, \dots, 0) = \vec{0}_{\mathbb{F}}$
since $T(z_1, \dots, z_m) = z_1 v_1 + \dots + z_m v_m$
the only $T(\vec{z}) = 0_V$ is $\vec{0}_{\mathbb{F}}$
$\Rightarrow \ker T = \vec{0}_{\mathbb{F}}$
T is inj. iff $\ker T = \{\vec{0}_{\mathbb{F}}\}$,
$\Rightarrow$ T is inj.
($\Leftarrow$) if T inj, $\ker T = \{\vec{0}_{\mathbb{F}}\}$
s.t. $T(\vec{z}) = 0 = z_1 v_1 + \dots + z_m v_m$
only for $z_1 = \dots = z_m = 0$,
since $\vec{0}_{\mathbb{F}} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}$,
thus the only $\vec{z}$ that solves
$\sum_{i=1}^{m} z_i v_i = 0$ is $z_1 = \dots = z_m = 0$, thus $v_1, \dots, v_m$ lin ind.

6. Proof: rank one means an outer product. Let A be a nonzero $m \times n$ matrix. Prove that $\operatorname{rank} A = 1$ if and only if there exist vectors $c \in \mathbb{F}^m$ and $d \in \mathbb{F}^n$ such that

$$A = cd^{\mathsf{T}}.$$

Unsure

Equivalently, prove that $A_{jk} = c_j d_k$ for all j, k.

$(\Rightarrow)$ if rank $A = 1$,

either $m = n = 1$, s.t. $A^{1\times 1} = a$

then $\exists\, c, d \in \mathbb{F}$ s.t. $a = cd$

for all $a \in \mathbb{F}$ by

closure of a field

or $m > 1$, $n > 1$,

s.t. $A^{m\times n} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \\ a_{m1} & & a_{mn} \end{bmatrix}$

since rank $A = 1$, w/o loss gen,

$\dim \operatorname{Col} A = 1$ s.t.

$\operatorname{Col} A = \operatorname{span}\{a_1, \ldots, a_n\}$

is 1 dim subspace of $\mathbb{F}^n$

ie a line in $\mathbb{F}^m$

thus $\exists\, c \in \mathbb{F}^m : c \neq 0$ s.t.

$\{c\}$ is basis for col A

$\Rightarrow \exists\, d_i \in \mathbb{F}$ s.t. $a_i = d_i c$

let $\vec{d} = \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix}$ where $d_i \in \mathbb{F}$

$\Rightarrow$ $A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix} = \begin{bmatrix} | & & | \\ d_1 c & \cdots & d_n c \\ | & & | \end{bmatrix}$

$A = \begin{bmatrix} c \\ \vdots \\ c \end{bmatrix} \begin{bmatrix} d_1 & \cdots & d_n \end{bmatrix} = \vec{c}\vec{d}^T$

thus $A_{jk} = (\vec{c}\vec{d}^T)_{jk} = c_j d_k$

$\Leftarrow$ if $A = \vec{c}\vec{d}^T$, $c \neq 0, d \neq 0$,

since $A \neq 0$

thus for at least one

$c_j \in \vec{c}$, $d_k \in \vec{d}^T$,

it must hold $c_j \neq 0, d_k \neq 0$

since $A_{jk} = c_j d_k \neq 0$

thus $\operatorname{rank}(A) \geq 1$

for any $x \in \mathbb{F}^n$,

$$Ax = (cd^T)x = c(d^Tx) = (d^Tx)c$$

since d^Tx is scalar in $\mathbb{F}$,

$$\operatorname{im} A \subseteq \operatorname{span}\{c\}$$

since $c \neq 0$,

$$\dim(\operatorname{span}\{c\}) = 1,$$

thus $\dim(\operatorname{im} A) \leq \dim(\operatorname{span}\{c\}) = 1$

s.t. $\dim(\operatorname{im} A) \leq 1$

since $A \neq 0$, $\dim(\operatorname{im} A) > 0$

thus $\dim(\operatorname{im} A) = 1$

& $\operatorname{rank} A = 1$

$\square$

III. Linear Maps, Invertibility, and Direct Sums

Checkpoint. The next problems are deliberately proof-oriented. They connect the calculations above to the basis-free statements used in lecture.

7. Let $A = (A_{jk})$ be an $n \times n$ matrix. Prove that the following are equivalent:

(a) the homogeneous system $Ax = 0$ has only the zero solution;

(b) for every $b \in \mathbb{F}^n$, the system $Ax = b$ has a solution.

Explain why this is the matrix form of the equivalence between injectivity and surjectivity for a linear operator on a finite-dimensional vector space.

$\Rightarrow$ if a), $Ax=0$ has only $\vec{x} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$ solution

thus for $A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix}$,

$\{a_1, \dots, a_n\}$ is lin ind set

s.t. $x_1 \vec{a}_1 + x_2 \vec{a}_2 + \dots + x_n \vec{a}_n = 0$

only for $x_1 = \dots = x_n = 0$

thus $\text{col } A = \text{span}\{a_1, \dots, a_n\}$

$\text{im } A = \mathbb{F}^n$

$\Rightarrow \forall \vec{b} \in \mathbb{F}^n, \exists \vec{x} \in \mathbb{F}^n$ s.t.

$x_1 \vec{a}_1 + \dots + x_n \vec{a}_n = \vec{b}$

thus $Ax = b$ has solution for any $b \in \mathbb{F}^n$

$\Leftarrow$ if $\forall b \in \mathbb{F}^n, \exists x \in \mathbb{F}^n$ s.t. $Ax = b$,

for a given x,

Ax is unique

thus for $b = 0$, there exists $x_0 \in \mathbb{F}^n$ s.t. $Ax = b = 0$

since $\operatorname{im} A = \operatorname{span}\{a_1, \ldots, a_n\}$
$= \mathbb{F}^n$ by the fact $Ax = b$ has solution for all b,

$\{a_1, \ldots, a_n\}$ lin. ind.

thus the sol. x_0 is unique

s.t. the only x satisfying $Ax = 0$ is $x_0 = 0$ $\square$

Explain why this is the matrix form of the equivalence between injectivity and surjectivity for a linear operator on a finite-dimensional vector space.

Lin operator implies $T(V,V)$

T inj. $\iff \ker T = \{0\}$ one to one

T surj $\iff \operatorname{im} T = V$ onto

if $A_T x = 0$ has only 0 solution, only x mapped by A_T to 0 is 0, such that $\ker A_T = \{0\}$

if $\forall b \in V, \exists x \in V$, s.t. $A_T x = b$, then entire codomain $V = \operatorname{span}\{b_1, \dots b_n\}$ is mapped by $A_T x$

s.t. $\operatorname{im} T = V \Rightarrow T$ surj.

8. Let $P \in \mathcal{L}(V)$ satisfy $P^2 = P$. Prove that

$$V = \operatorname{Ker} P \oplus \operatorname{Im} P.$$

Then, for an arbitrary $v \in V$, find explicit formulas for its component in $\operatorname{Ker} P$ and its component in $\operatorname{Im} P$.

Couldn't solve

$P : V \to V \qquad P^2 = P \implies Px = PPx$

$\operatorname{Ker} P = \{v \in V : Pv = 0\}$

$\operatorname{Im} P = \{y = Px \ \forall x \in V\}$

1. **Fixing the Image:** P acts as the identity operator on its own range.

$$\exists u \in V \text{ s.t. } w = Pu \implies Pw = P(Pu) = P^2 u = Pu = w$$

2. **Complementary Annihilation:** The identity operator decomposes as $I = (I - P) + P$, where:

$$P(I - P) = P - P^2 = P - P = 0$$

This guarantees that for any $v \in V$, the vector $(I - P)v$ lies in $\operatorname{Ker} P$.

8. Let $P \in \mathcal{L}(V)$ satisfy $P^2 = P$. Prove that

$$V = \operatorname{Ker} P \oplus \operatorname{Im} P.$$

Couldn't solve

Then, for an arbitrary $v \in V$, find explicit formulas for its component in $\operatorname{Ker} P$ and its component in $\operatorname{Im} P$.

$P: V \to V \qquad P^2 = P \implies Px = PPx$

$\operatorname{Ker} P = \{v \in V : Pv = 0\}$

$\operatorname{Im} P = \{y = Px \ \forall x \in V\}$

P acts like a projection, s.t.

$\exists\, u \in V$ s.t. $Pu = w$

$\implies PPu = Pw$

$Pu = Pw$

$w = Pw \implies \forall w$ in image of P, P acts as I

Compl. Annihilation

$\forall v \in V, (I - P)v = 0$

3. Formal Proof: $V = \text{Ker } P \oplus \text{Im } P$

To establish that V is the internal direct sum of $\text{Ker } P$ and $\text{Im } P$, we must show:

1. $\text{Ker } P \cap \text{Im } P = \{0\}$
2. $V = \text{Ker } P + \text{Im } P$

Part 1: Trivial Intersection ($\text{Ker } P \cap \text{Im } P = \{0\}$)

Let $v \in \text{Ker } P \cap \text{Im } P$.

- Since $v \in \text{Ker } P$, $Pv = 0$.
- Since $v \in \text{Im } P$, there exists $u \in V$ such that $v = Pu$.

Applying P to v:

$$v = Pu = P^2 u = P(Pu) = Pv$$

Since $Pv = 0$, it follows directly that:

$$v = 0$$

Hence, $\text{Ker } P \cap \text{Im } P = \{0\}$.

Claim:

$V = \text{Ker } P \oplus \text{Im } P$

To prove:

1) $\text{Ker } P \cap \text{Im } P = \{0\}$

2) $V = \text{Ker } P + \text{Im } P$

1) let $v \in \text{Ker } P \cap \text{Im } P$

s.t. $v \in \text{Ker } P$

$\Rightarrow Pv = 0$ (*)

since $v \in \text{Im } P$,

$\exists\, u \in V$ s.t.

$v = Pu$

$\Rightarrow v = Pu = P^2 u = P(Pu) = Pv$

by (*), $Pv = 0 \Rightarrow v = 0$

thus $\text{Ker } P \cap \text{Im } P = \{0\}$

Part 2: Spanning the Space ($V = \text{Ker } P + \text{Im } P$)

Let $v \in V$ be arbitrary. Decompose v algebraically:

$$v = (v - Pv) + Pv$$

We verify the subspace membership for each term:

- **For Pv:** By definition, $Pv \in \text{Im } P$.
- **For $v - Pv$:** Applying the linear operator P:

$$P(v - Pv) = Pv - P^2 v = Pv - Pv = 0$$

Therefore, $(v - Pv) \in \text{Ker } P$.

Because every $v \in V$ can be expressed as the sum of an element in $\text{Ker } P$ and an element in $\text{Im } P$, we have $V = \text{Ker } P + \text{Im } P$.

Combining Parts 1 and 2 completes the proof:

$$V = \text{Ker } P \oplus \text{Im } P$$

2) $\forall v \in V$

$V = V - Pv + Pv$ (*)

since $Pv - Pv = 0$

split into $v - Pv$ & Pv

i) by def $Pv \in \text{Im} P$

ii) apply P to $v - P$

$P(v - Pv) = Pv - P^2 v$
$= Pv - Pv = 0$
$\Rightarrow P(v - Pv) = 0$
& $(v - Pv) \in \text{Ker} P$

Since $\forall v \in V$, v is sum of elem of $\text{Im} P$ & $\text{Ker} P$, $V = \text{Ker} P + \text{Im} P$

Thus by 1 & 2,

$V = \text{Ker } P \oplus \text{Im} P$

IV. Eigenvalues and How to Diagonalize a Matrix

Mini-lecture. For a square matrix A, an eigenvalue λ is a scalar for which

$$(A - \lambda I)v = 0$$

has a nonzero solution. For a 2×2 matrix, the computation can be organized by the characteristic equation

$$\det(A - \lambda I) = 0, \qquad \det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Once λ is known, find its eigenvectors by row-reducing $A - \lambda I$. If an $n \times n$ matrix has n linearly independent eigenvectors $v_1, \ldots, v_n$, place them in the columns of

$$X = (v_1 \ \cdots \ v_n)$$

and put the corresponding eigenvalues on the diagonal of D. Then

$$AX = XD, \qquad \text{hence} \qquad A = XDX^{-1}.$$

This is the computational meaning of diagonalization.

Worked example. Let

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

The characteristic equation is

$$\det\begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix} = (4-\lambda)(3-\lambda) - 2 = (\lambda - 5)(\lambda - 2) = 0.$$

Thus the eigenvalues are 5 and 2.

For $\lambda = 5$,

$$(A - 5I)v = 0 \quad \Longrightarrow \quad v \in \operatorname{span}\{(1,1)^{\mathsf{T}}\}.$$

For $\lambda = 2$,

$$(A - 2I)v = 0 \quad \Longrightarrow \quad v \in \operatorname{span}\{(1,-2)^{\mathsf{T}}\}.$$

Choose

$$X = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, \qquad D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}.$$

The columns of X are independent, so X is invertible, and $AX = XD$. Therefore

$$A = XDX^{-1}.$$

This is the pattern to imitate in the next computational problems.

9. First eigenvalue calculation. Define $T \in \mathcal{L}(\mathbb{F}^2)$ by

$$T(w,z) = (z,w).$$

(a) Write the matrix of T in the standard basis.

(b) Find all eigenvalues.

(c) For each eigenvalue, solve the corresponding homogeneous system to find the eigenspace.

(d) Choose an eigenbasis and write the diagonal matrix of T in that basis.

a) swaps v_1 & v_2 of $\vec{v}$,

$$M_T = A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} z \\ w \end{bmatrix}$$

b) $|A - \lambda I| = 0$ $\quad \left| \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$

$$\left| \begin{bmatrix} \lambda & 1 \\ 1 & \lambda \end{bmatrix} \right| = 0$$

$$\lambda^2 - 1 = 0 \Rightarrow (\lambda+1)(\lambda-1)$$

$$\lambda_1 = 1, \quad \lambda_2 = -1$$

c) 1) $(A - \lambda_1 I)x = 0$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \begin{aligned} -x_1 + x_2 &= 0 \\ x_1 - x_2 &= 0 \end{aligned}$$

$$x_1 = x_2$$

$$V_1 = \operatorname{span}\{[1,1]^T\}$$

2) $(A - \lambda_2 I)x = 0$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \begin{aligned} x_1 + x_2 &= 0 \\ x_1 + x_2 &= 0 \end{aligned} \qquad x = -x_2$$

$$V_2 = \operatorname{span}\{[1,-1]^T\}$$

d) Choose $X = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ $\det X = -1-1 = -2 \neq 0$

$D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ X is invert.

$$AX = XD$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

thus $D = X^{-1}AX$

w/ $X^{-1} = \frac{1}{-2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$

$= \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

10. A matrix that does not have enough eigenvectors. Define $T \in \mathcal{L}(\mathbb{F}^3)$ by

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3).$$

(a) Write the standard matrix A of T.

(b) Find all eigenvalues and their eigenspaces.

(c) Count the number of linearly independent eigenvectors that can be obtained.

(d) Decide whether A can be diagonalized, and explain the answer using your calculation rather than quoting a theorem without verification.

a)
$$\underset{A}{\overset{3\times 3}{\begin{bmatrix}0&2&0\\0&0&0\\0&0&5\end{bmatrix}}}\underset{\vec{z}}{\overset{3\times 1}{\begin{bmatrix}z_1\\z_2\\z_3\end{bmatrix}}}=\underset{\vec{z}'}{\overset{3\times 1}{\begin{bmatrix}2z_2\\0\\5z_3\end{bmatrix}}}$$

b) $\det(A-\lambda I)=0$

$$\left|\begin{bmatrix}-\lambda&2&0\\0&-\lambda&0\\0&0&5-\lambda\end{bmatrix}\right| = 0$$

$$-\lambda\begin{vmatrix}-\lambda&0\\0&5-\lambda\end{vmatrix} - 2\begin{vmatrix}0&0\\0&-\lambda\end{vmatrix} + 0\begin{vmatrix}0&-\lambda\\0&0\end{vmatrix}$$

$-\lambda(-\lambda(5-\lambda))$ $\quad 0 \quad 0$

$-\lambda(\lambda^2-5\lambda)$

$5\lambda^2-\lambda^3$

$=5\lambda^2-\lambda^3=0$

$\lambda^2(5-\lambda)=0$

$\lambda_1=5,\ \lambda_2=0,\ \lambda_3=0$

c) multiplicity of 2 for $\lambda_2=\lambda_3=0$

so only 2 lin ind ev

d) $(A-\lambda I)v=0$

$$\begin{bmatrix}-5&2&0\\0&-5&0\\0&0&0\end{bmatrix}v=0 \qquad \cancel{-5v_1+2v_2=0}$$

$$\begin{bmatrix}0&2&0\\0&0&0\\0&0&5\end{bmatrix}v=0 \qquad 2v_2=0,\ 5v_3=0,\ v_1 \text{ free}$$

$v_2 = 0 \Rightarrow v_1=0$, v_3 free $\begin{bmatrix}0\\0\\v_3\end{bmatrix}$ $\qquad \begin{bmatrix}v_1\\0\\0\end{bmatrix}$ for x_2 and x_3

multiplicity of x_2, x_3 $\Rightarrow$ X not lin ind

X non-inv, A not diag.

11. Diagonalization and powers. Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}.$$

(a) Find the eigenvalues of A.

(b) Find one eigenvector for each eigenvalue.

(c) Form the eigenvector matrix X and the diagonal matrix D.

(d) Verify directly that $AX = XD$, and hence that $A = XDX^{-1}$.

(e) Use

$$A^k = XD^kX^{-1}$$

to derive a closed-form expression for A^k.

a) $\det \begin{pmatrix} 1-\lambda & 2 \\ 0 & 3-\lambda \end{pmatrix} = 0$

$(1-\lambda)(3-\lambda) - (2)(0) = 0$

$3 - 3\lambda - \lambda + \lambda^2 = 0$

$\lambda^2 - 4\lambda + 3 = 0$

$(\lambda - 1)(\lambda - 3) = 0$

$\lambda_1 = 1 \quad \lambda_2 = 3$

b)

λ_1) $\begin{pmatrix} 1-1 & 2 \\ 0 & 3-1 \end{pmatrix} v = 0$

$\begin{pmatrix} 0 & 2 \\ 0 & -2 \end{pmatrix} v = 0$ $\quad v_1$ free

$2v_2 = 0$

$-2v_2 = 0$

$v_2 = 0 \Rightarrow X_1 = \begin{bmatrix} v_1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

λ_2) $\begin{pmatrix} 1-3 & 2 \\ 0 & 3-3 \end{pmatrix} v = 0$

$\begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix} v = 0$

$-2v_1 + 2v_2 = 0$

$v_1 = v_2 \Rightarrow X_2 = \begin{bmatrix} v_1 \\ v_1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

c) $X = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $\quad D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

d) $AX = XD$

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 3 \end{bmatrix} \quad \checkmark$$

$\det X = 1$, X^{-1} exists

e) $A^k = X D^k X^{-1}$ $\qquad X^{-1} = \frac{1}{1}\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

$$X X^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^k = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1^k & 0 \\ 0 & 3^k \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1^k & 3^k \\ 0 & 3^k \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$-1^k + 3^k$

$$= \begin{bmatrix} 1^k & 2^k \\ 0 & 3^k \end{bmatrix} = A^k$$

12. Proof: similarity preserves eigenvalues. Let $T \in \mathcal{L}(V)$ and let $S \in \mathcal{L}(V)$ be invertible.

(a) Prove that T and $S^{-1}TS$ have the same eigenvalues.

(b) If v is an eigenvector of T with eigenvalue λ, identify an eigenvector of $S^{-1}TS$ with the same eigenvalue.

Explain why changing basis changes the matrix representing a linear operator but not its eigenvalues.

a) let $S^{-1}TS = B$

$\exists\, v_T, v_B \in V$ eigenvectors w/ $v_T \neq 0, v_B \neq 0$,

s.t. $Tv_T = \lambda_T v_T$ & $Bv_B = \lambda_B v_B$

Since S invertible, S surj. & inj.

s.t. $Su = v_T$ for some $u \in V$ (surj)

$\Rightarrow u = S^{-1} v_T$

Since $v_T \neq 0$, $u = S^{-1} v_T \neq 0$ (inj.)

apply $S^{-1}TS$ to u:

$S^{-1}TSu = S^{-1}T(Su) = S^{-1}T v_T$

$= S^{-1}\lambda_T v_T = \lambda_T S^{-1} v_T = \lambda_T u$

$\Rightarrow S^{-1}TSu = \lambda_T u$

$Bu = \lambda_T u \Rightarrow$ λ_T is eigenvalue for $B = S^{-1}TS$

Direction 1: If λ is an eigenvalue of T, then λ is an eigenvalue of $S^{-1}TS$

1. Let λ be an eigenvalue of T. By definition, there exists a non-zero vector $v \in V$ $(v \neq 0)$ such that:

   $$Tv = \lambda v$$

2. Since S is invertible, S is surjective, so there exists a vector $u \in V$ such that $Su = v$ (or equivalently, define $u = S^{-1}v$).
3. Because S is injective and $v \neq 0$, it follows that:

   $$u = S^{-1}v \neq 0$$

4. Apply the operator $S^{-1}TS$ to the vector u:

   $$(S^{-1}TS)u = S^{-1}T(Su) = S^{-1}(Tv)$$

5. Substitute $Tv = \lambda v$:

   $$S^{-1}(Tv) = S^{-1}(\lambda v) = \lambda(S^{-1}v) = \lambda u$$

6. Therefore, $(S^{-1}TS)u = \lambda u$ with $u \neq 0$, proving λ is an eigenvalue of $S^{-1}TS$ with corresponding eigenvector $u = S^{-1}v$.

Direction 2: If λ is an eigenvalue of $S^{-1}TS$, then λ is an eigenvalue of T

- Let $(S^{-1}TS)w = \lambda w$ with $w \neq 0$.
- Left-multiply by S:

$$S(S^{-1}TS)w = S(\lambda w) \Longrightarrow T(Sw) = \lambda(Sw)$$

- Setting $z = Sw$, since S is invertible and $w \neq 0$, $z \neq 0$.
- Hence $Tz = \lambda z$ with $z \neq 0$, proving λ is an eigenvalue of T.

$$\text{spec }(T) = \text{ spec }(S^{-1}TS)$$

for λ_B, $Bv_B = \lambda_B v_B$

$S^{-1}TSv_B = \lambda_B v_B$ for $v_B \neq 0$

left multiply by S:

$S(S^{-1}TS)v_B = S(\lambda_B v_B)$ let $Sv_B = z$

$TSv_B = \lambda_B S v_B$ $\Rightarrow Tz = \lambda_B z$, thus λ_B is eigenvalue of T

Since λ_B is ev of T & λ_T is ev of B, both have same evs

(b) If v is an eigenvector of T with eigenvalue λ, identify an eigenvector of $S^{-1}TS$ with the same eigenvalue.

for V_T eigenvector of T & λ_T

$T v_T = \lambda_T v_T$

find v_B for $S^{-1}TS = B$ s.t.

$B v_B = \lambda_T v_B$

by a) all λ_T are eigenvalues of B s.t. $\exists\, v_B$ associated w/ λ_T s.t. $B v_B = \lambda_T v_B$

thus v_B shares λ_T w/ V_T

Explain why changing basis changes the matrix representing a linear operator but not its eigenvalues.

Matrix composed of vectors w/ coord. in given basis, changing basis adjusts coord but maintains inherent position vectors, so degree of scaling in the principal directions is invariant

Just for studying purposes:

A linear operator $T : V \to V$ is a coordinate-free geometric mapping on a vector space V. A matrix representation of T depends explicitly on the choice of basis used to parametrize V, whereas the eigenvalues of T are intrinsic algebraic and geometric properties of the operator itself.

1. Effect of a Change of Basis on the Matrix Representation

Let V be an n-dimensional vector space over a field $\mathbb{F}$, and let $\mathscr{B} = \{v_1, \ldots, v_n\}$ and $\mathscr{B}' = \{v'_1, \ldots, v'_n\}$ be two distinct ordered bases for V.

- The matrix representation of T with respect to $\mathscr{B}$, denoted $[T]_{\mathscr{B}} = A \in \mathbb{F}^{n\times n}$, is defined such that for any vector $x \in V$:

$$[T(x)]_{\mathscr{B}} = A[x]_{\mathscr{B}}$$

  where the j-th column of A contains the coordinates of $T(v_j)$ relative to $\mathscr{B}$.

- Let $P = [I]_{\mathscr{B}}^{\mathscr{B}'} \in \mathrm{GL}_n(\mathbb{F})$ be the invertible change-of-basis matrix converting $\mathscr{B}'$-coordinates to $\mathscr{B}$-coordinates:

$$[x]_{\mathscr{B}} = P[x]_{\mathscr{B}'}$$

- Expressing the action of T in the $\mathscr{B}'$ basis yields:

$$[T(x)]_{\mathscr{B}'} = P^{-1}[T(x)]_{\mathscr{B}} = P^{-1}A[x]_{\mathscr{B}} = P^{-1}AP[x]_{\mathscr{B}'}$$

- Consequently, the matrix representation with respect to $\mathscr{B}'$ is:

$$B = [T]_{\mathscr{B}'} = P^{-1}AP$$

Because $P \neq I$ in general, $B \neq A$. A change of basis modifies the matrix representation via a similarity transformation.

- **Coordinate-Free Geometric Characterization:**
  A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of T if and only if there exists a non-zero vector $v \in V$ such that:

$$(T - \lambda I)v = 0$$

  This definition relies solely on the action of T on the underlying vector space V. Since vectors and linear operators exist independently of coordinate frames, the existence of such a scalar λ and invariant subspace $\ker(T - \lambda I)$ does not depend on the choice of basis.

- **Invariance of the Characteristic Polynomial:**
  Let $A = [T]_{\mathscr{B}}$ and $B = P^{-1}AP = [T]_{\mathscr{B}'}$. The characteristic polynomial $p_B(\lambda)$ satisfies:

$$\det(\lambda I - B) = \det(\lambda I - P^{-1}AP) = \det(P^{-1}(\lambda I - A)P)$$

  Applying the multiplicativity of the determinant:

$$P) = \frac{1}{\det(P)} \det(\lambda I - A) \det(P) = \det(\lambda I - A) = p_A(\lambda)$$

  Because A and B share identical characteristic polynomials, their roots—the eigenvalues λ_i along with their algebraic multiplicities—are strictly identical.

- **Explicit Eigenvector Coordinate Transformation:**
  If $x \in \mathbb{F}^n \setminus \{0\}$ is an eigenvector of A with eigenvalue λ, then $Ax = \lambda x$. Transforming x into the coordinate system of $\mathscr{B}'$ via $y = P^{-1}x \neq 0$:

$$By = (P^{-1}AP)(P^{-1}x) = P^{-1}Ax = P^{-1}(\lambda x) = \lambda(P^{-1}x)$$

  Thus, y is an eigenvector of B corresponding to the identical eigenvalue λ, preserving both the eigenvalues and the geometric multiplicity $\dim(\ker(A - \lambda I)) = \dim(\ker(B - \lambda I))$.

V. Orthogonality, Gram–Schmidt, and Projection

Mini-lecture. Given linearly independent vectors $v_1, \dots, v_m$, the Gram–Schmidt procedure constructs an orthonormal list spanning the same successive subspaces. For the first two vectors,

$$e_1 = \frac{v_1}{\|v_1\|}, \qquad u_2 = v_2 - \langle v_2, e_1 \rangle e_1, \qquad e_2 = \frac{u_2}{\|u_2\|}.$$

If $e_1, \dots, e_m$ is an orthonormal basis of a subspace U, then the orthogonal projection of b onto U is

$$P_U b = \sum_{j=1}^{m} \langle b, e_j \rangle e_j.$$

The residual $b - P_U b$ is orthogonal to every vector in U, which is why $P_U b$ is the closest point in U to b.

Worked example. Apply Gram–Schmidt to $v_1 = (1, 1)$ and $v_2 = (1, 0)$ in $\mathbb{R}^2$:

$$e_1 = \frac{1}{\sqrt{2}}(1, 1),$$

$$u_2 = (1, 0) - \left\langle (1, 0), \frac{1}{\sqrt{2}}(1, 1) \right\rangle \frac{1}{\sqrt{2}}(1, 1) = \frac{1}{2}(1, -1),$$

so

$$e_2 = \frac{1}{\sqrt{2}}(1, -1).$$

Thus e_1, e_2 is an orthonormal basis spanning the same space as v_1, v_2.

13. On $\mathcal{P}_2(\mathbb{R})$ define

$$\langle p, q\rangle = \int_0^1 p(x)q(x)\,dx.$$

Apply the Gram–Schmidt procedure to

$$1, x, x^2$$

to produce an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$. Show every subtraction and normalization step.

$$\vec{X} = \begin{bmatrix} 1 \\ x \\ x^2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \qquad e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{\langle v_1, v_1\rangle}} =$$

$$\frac{1}{\sqrt{\int_0^1 1 \cdot 1\,dx}} = \frac{1}{\sqrt{1}} = 1$$

$$u_2 = v_2 - \langle v_2, e_1\rangle e_1 = x - \langle x, 1\rangle 1$$

$$= x - \int_0^1 x \cdot 1\,dx$$

$$= x - \left[\tfrac{1}{2}x^2\right]_0^1 = x - \tfrac{1}{2}$$

$$e_2 = \frac{u_2}{\|u_2\|} = \frac{x - 1/2}{\sqrt{\langle x-\frac{1}{2}, x-\frac{1}{2}\rangle}} = \frac{x-1/2}{\sqrt{\int_0^1 (x-\frac{1}{2})(x-\frac{1}{2})dx}} = \frac{x-1/2}{\sqrt{\int_0^1 x^2 - x + \frac{1}{4}\,dx}}$$

$$\frac{x - 1/2}{\sqrt{\left[\frac{1}{3}x^3 - \frac{1}{2}x^2 + \frac{1}{4}x\right]_0^1}} = \frac{x - 1/2}{\sqrt{\left(\frac{1}{3} - \frac{1}{2} + \frac{1}{4}\right)}} = \frac{x - 1/2}{\sqrt{1/12}} = \frac{1}{2\sqrt{3}}\left(x - \frac{1}{2}\right)$$

$$u_3 = v_3 - \langle v_3, e_2\rangle e_2 - \langle v_3, e_1\rangle e_1$$

$$x^2 - \left\langle x^2, \frac{x-1/2}{\sqrt{1/12}}\right\rangle \frac{x-1/2}{\sqrt{1/12}} - \langle x^2, 1\rangle 1$$

$$x^2 - \int_0^1 x^2\left(\frac{x-1/2}{\sqrt{1/12}}\right)dx \left(\frac{x-1/2}{\sqrt{1/12}}\right) - \int_0^1 x^2\,dx$$

$$x^2 - \int_0^1 x^2\left(\frac{x-1/2}{\sqrt{1/12}}\right)dx\left(\frac{x-1/2}{\sqrt{1/12}}\right) - \int_0^1 x^2\,dx$$

$$x^2 - \frac{x-1/2}{1/12}\int_0^1 x^3 - \frac{1}{2}x^2\,dx - \left[\frac{1}{3}x^3\right]_0^1$$

$$x^2 - 12\left(x-\frac{1}{2}\right)\left[1/4x^4 - \frac{1}{6}x^3\right]_0^1 \quad - \left(\frac{1}{3}\right)$$

$$x^2 - (12x-6)\left(\frac{1}{4}-\frac{1}{6}\right) \quad - \frac{1}{3} \qquad\qquad \frac{3}{2} - 1 - \frac{1}{3}$$

$$x^2 - \left(3x - 2x - \frac{3}{2} + 1\right) - \frac{1}{3} = x^2 - x + \frac{1}{6} = u_3$$

$$e_3 = \frac{u_3}{\|u_3\|} = \frac{x^2 - x + 1/6}{\sqrt{\langle x^2 - x + \frac{1}{6}, x^2 - x + \frac{1}{6}\rangle}}$$

$$= \frac{x^2 - x + 1/6}{\sqrt{\int_0^1 (x^2 - x + \frac{1}{6})(x^2 - x + \frac{1}{6})\,dx}}$$

$$= \frac{x^2 - x + \frac{1}{6}}{\sqrt{\int_0^1 x^4 - x^3 + \frac{1}{6}x^2 - x^3 + x^2 - \frac{1}{6}x + \frac{1}{6}x^2 - \frac{1}{6x} + \frac{1}{36}}}$$

$$x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + 1/36$$

$$\left[\frac{1}{5}x^5 - \frac{1}{2}x^4 + \frac{4}{9}x^3 - \frac{1}{6}x^2 + \frac{1}{36}x\right]_0^1$$

$$\frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} =$$

$$= \frac{x^2 - x + 1/6}{\sqrt{1/180}} = 6\sqrt{5}\left(x^2 - x + \frac{1}{6}\right) = e_3$$

$$= \sqrt{5}\left(6x^2 - 6x + 1\right)$$

Orthonormal Basis $= \{e_1, e_2, e_3\} =$

$$\left\{1, \frac{1}{2\sqrt{3}}\left(x - \frac{1}{2}\right), \sqrt{5}\left(6x^2 - 6x + 1\right)\right\}$$

14. In $\mathbb{R}^4$, let

$$U = \operatorname{span}\{(1,1,0,0),(1,1,1,2)\}.$$

Find the vector $u \in U$ that minimizes

$$\|u - (1,2,3,4)\|.$$

You may first orthonormalize a basis of U. After finding u, verify explicitly that the residual $(1,2,3,4) - u$ is orthogonal to both spanning vectors of U.

$$e_1 = \frac{v_1}{\|v_1\|}, \qquad u_2 = v_2 - \langle v_2, e_1\rangle e_1, \qquad e_2 = \frac{u_2}{\|u_2\|}.$$

a) GS on

$$V_1 = \begin{bmatrix}1\\1\\0\\0\end{bmatrix}, \quad V_2 = \begin{bmatrix}1\\1\\1\\2\end{bmatrix}$$

$$e_1 = \frac{V_1}{\|V_1\|_2} = \frac{(1\ 1\ 0\ 0)^T}{\sqrt{1^2+1^2+0^2+0^2}} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\\0\end{bmatrix} = \begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\\0\\0\end{bmatrix}$$

$$U_2 = V_2 - \langle V_2, e_1\rangle e_1$$

$$\begin{bmatrix}1\\1\\1\\2\end{bmatrix} - \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 0 + 0\right)\begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\\0\\0\end{bmatrix} = \begin{bmatrix}1\\1\\1\\2\end{bmatrix} - \frac{2}{\sqrt{2}}\begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\\0\\0\end{bmatrix} = \begin{bmatrix}1-1\\1-1\\1\\2\end{bmatrix} = \begin{bmatrix}0\\0\\1\\2\end{bmatrix}$$

$$e_2 = \frac{U_2}{\|U_2\|} = \frac{U_2}{\sqrt{0+0+1+4}} = \frac{1}{\sqrt{5}}\begin{bmatrix}0\\0\\1\\2\end{bmatrix} = \begin{bmatrix}0\\0\\1/\sqrt{5}\\2/\sqrt{5}\end{bmatrix}$$

$$B_e = \left\{\begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1/\sqrt{5}\\2/\sqrt{5}\end{bmatrix}\right\}$$

let $v = (1,2,3,4)$ in original basis U

$$\min u = P_U(v) = \langle v, e_1\rangle e_1 + \langle v, e_2\rangle e_2$$

by projecting v w/ B_e

$$P_u(v) = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2$$

$$v = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix} \quad e_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$\frac{3}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} + \frac{11}{\sqrt{5}} \begin{bmatrix} 0 \\ 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$\begin{bmatrix} 3/2 \\ 3/2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 11/5 \\ 22/5 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 3/2 \\ 11/5 \\ 22/5 \end{bmatrix} = u$$

Verify

$$r = v - u = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 3/2 \\ 3/2 \\ 11/5 \\ 22/5 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \\ 4/5 \\ -2/5 \end{bmatrix}$$

$r \perp v_1 = (1, 1, 0, 0)^T$?

$$\langle r, v_1 \rangle = -1/2 + 1/2 + 0 + 0 = 0 \quad \checkmark$$

$r \perp v_2 = (1, 1, 1, 2)^T$?

$$\langle r, v_2 \rangle = -\frac{1}{2} + \frac{1}{2} + \frac{4}{5} - \frac{4}{5} = 0 \quad \checkmark$$

$$u = (3/2, 3/2, 11/5, 22/5)^T$$

15. Proof: orthogonality as a minimization condition. Suppose $u, v \in V$. Prove that

$$\langle u, v\rangle = 0 \quad \iff \quad \|u\| \le \|u + av\| \quad \text{for every } a \in \mathbb{F}.$$

Interpret the result as the simplest version of the least-squares first-order condition.

$\Rightarrow$ if $\langle u, v \rangle = 0$,

let $a \in \mathbb{F}$ be some scalar,

by induced norm $\|u\|^2 = \langle u, u \rangle$

we know $\|u+av\|^2 = \langle u+av, u+av \rangle$

$= \langle u, u+av \rangle + \langle av, u+av \rangle$

by lin in first arg

$= \langle u, u \rangle + \bar{a}\langle u, v \rangle + a\langle v, u \rangle + a\bar{a}\langle v, v \rangle$

by conj. lin in 2nd

thus

$\|u+av\|^2 = \langle u, u \rangle + a\langle v, u \rangle + \bar{a}\langle v, u \rangle + |a|^2\langle v, v \rangle$

$\langle u, u \rangle = \|u\|^2$ & $\langle v, v \rangle = \|v\|^2$

$\|u+av\|^2 = \|u\|^2 + a\langle v, u \rangle + \bar{a}\langle v, u \rangle + |a|^2\|v\|^2$

since $\langle u, v \rangle = 0$,

$\|u+av\|^2 = \|u\|^2 + |a|^2\|v\|^2$

since $|a|^2\|v\|^2 \geq 0$,

$\|u+av\|^2 \geq \|u\|^2$

sqrt both sides

$\|u+av\| \geq \|u\|$ for all $a \in \mathbb{F}$

$\Leftarrow$ if $\|u\| \leq \|u+av\|$ for $\forall a \in \mathbb{F}$,

square both sides

$$\|u\|^2 \leq \|u+av\|^2$$

$$\|u\|^2 \leq \|u\|^2 + 2\bar{a}\langle u, v\rangle + |a|^2\|v\|^2$$

$$\Rightarrow \quad 0 \leq 2\bar{a}\langle u, v\rangle + |a|^2\|v\|^2$$

if $v = 0$, $\langle u, v\rangle = 0$ b/c $u_1(0) + \dots + u_n 0 = 0$

for $v \neq 0$, $0 < |a|^2\|v\|^2$

$$\Rightarrow 0 > -|a|\|v\|^2$$

$$-|a|^2\|v\|^2 \leq 2\bar{a}\langle u, v\rangle$$

$$-|a|^2\|v\|^2 < 0 \leq 2\bar{a}\langle u, v\rangle$$

???

$$2\bar{a}\langle u, v\rangle \leq 0 \leq 2\bar{a}\langle u, v\rangle$$

thus $\langle u, v\rangle = 0$

$\square$

2. **Non-Trivial Case ($v \neq 0$):**
   Assume $v \neq 0$, so $\|v\|^2 > 0$. Expanding the squared norm for any $a \in \mathbb{F}$:

$$\|u + av\|^2 = \langle u + av, u + av \rangle = \|u\|^2 + a\langle v, u \rangle + \bar{a}\langle u, v \rangle + |a|^2\|v\|^2$$

   Using $\langle v, u \rangle = \overline{\langle u, v \rangle}$, this becomes:

$$\|u + av\|^2 = \|u\|^2 + 2\,\mathrm{Re}\,(\bar{a}\langle u, v \rangle) + |a|^2\|v\|^2$$

3. **The Missing Step (Choosing a):**
   Since the inequality $\|u\| \leq \|u + av\|$ holds for *all* $a \in \mathbb{F}$, choose the specific scalar:

$$a = -\frac{\langle u, v \rangle}{\|v\|^2}$$

had to look up solution

4. **Substitution & Simplification:**
   Substitute this choice of a into the expansion:

$$\|u + av\|^2 = \|u\|^2 - \frac{\langle u, v \rangle}{\|v\|^2}\langle v, u \rangle - \frac{\overline{\langle u, v \rangle}}{\|v\|^2}\langle u, v \rangle + \frac{|\langle u, v \rangle|^2}{\|v\|^4}\|v\|^2$$

$$\|u + av\|^2 = \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2} - \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \frac{|\langle u, v \rangle|^2}{\|v\|^2}$$

$$\|u + av\|^2 = \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2}$$

5. **Conclusion:**
   The hypothesis $\|u\|^2 \leq \|u + av\|^2$ yields:

$$\|u\|^2 \leq \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2} \implies \frac{|\langle u, v \rangle|^2}{\|v\|^2} \leq 0$$

   Because $\|v\|^2 > 0$ and $|\langle u, v \rangle|^2 \geq 0$, it must be that:

$$|\langle u, v \rangle|^2 = 0 \implies \langle u, v \rangle = 0$$

Least-Squares First-Order Condition Interpretation

In the optimization problem:

$$\min_{a \in \mathbb{F}} \|u + av\|^2$$

the objective function $f(a) = \|u\|^2 + 2\,\mathrm{Re}\,(\bar{a}\langle u, v\rangle) + |a|^2\|v\|^2$ is strictly convex with respect to a. The point $a = 0$ is the global minimizer if and only if the directional derivative (gradient) at 0 vanishes:

$$\nabla f(0) = 2\langle u, v\rangle = 0 \iff \langle u, v\rangle = 0$$

This expresses the classic projection theorem: the error vector u has minimal norm if and only if it is orthogonal to the subspace spanned by v.

Optional Exploratory Problem

16. What failure of diagonalization looks like. Let $T \in \mathcal{L}(\mathbb{C}^2)$ be defined by

$$T(w, z) = (z, 0).$$

(a) Find all eigenvalues and eigenspaces of T.

(b) Is there a basis of $\mathbb{C}^2$ consisting of eigenvectors? Explain.

(c) Compute T^2.

(d) Find v such that $Tv \neq 0$ but $T^2 v = 0$.

(e) Show that (Tv, v) is a basis and write the matrix of T in this basis.

(f) The resulting matrix is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Explain informally what extra information this matrix records that ordinary eigenvectors miss.

a)

$$T(w, z) = (z, 0) \Rightarrow A \begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} z \\ 0 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Set $\det(A - \lambda I) = 0 \Rightarrow \det \begin{bmatrix} -\lambda & 1 \\ 0 & -\lambda \end{bmatrix} = 0$

$$\lambda^2 - 0 = 0 \Rightarrow \lambda^2 = 0$$

$$\lambda_1 = \lambda_2 = 0$$

$Av = \lambda v = 0$

$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow$ v_1 free, $v_2 = 0$ $\Rightarrow v = \begin{bmatrix} t \\ 0 \end{bmatrix}$

eigenspace = span$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$

b) No, since only eigenvectors are of form $\begin{bmatrix} t \\ 0 \end{bmatrix}$, they can only span 1 dimension, so cannot form a basis for $\mathbb{C}^2$

c) $T^2 = T(T(w,z)) = T(z,0) = (0,0)$

$$AA\begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} w \\ z \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} z \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(d) Find v such that $Tv \neq 0$ but $T^2v = 0$.

(e) Show that (Tv, v) is a basis and write the matrix of T in this basis.

d) $T(v_1, v_2) \neq 0 \quad T^2(v_1, v_2) = 0$ applies for any $v_2 \neq 0$

e) (Tv, v) for $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ w/ $v_2 \neq 0$

$= \left\{\begin{bmatrix} v_2 \\ 0 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right\}$ lin ind & spans $\mathbb{C}^2$

$\forall c_1, c_2 \in \mathbb{C}^2$, $a_1 v_2 + a_2 v_1 = c_1$, $a_2 v_2 = c_2$

$\exists a_1, a_2 \Rightarrow$ by closure of $\mathbb{C}$

since $a_1\begin{bmatrix} v_2 \\ 0 \end{bmatrix} + a_2\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

$= \begin{bmatrix} a_1 v_2 \\ 0 \end{bmatrix} + \begin{bmatrix} a_2 v_1 \\ a_2 v_2 \end{bmatrix}$

$= \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

only if $a_2 v_2 = 0 \Rightarrow a_2 = 0$

& $a_1 v_2 = 0 \Rightarrow a_1 = 0$

e) Matrix of T in $\left\{ \begin{bmatrix} v_2 \\ 0 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\}$

s.t. $T(w, z) = (z, 0)$

$$= \left[0 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, 1 \begin{bmatrix} v_2 \\ 0 \end{bmatrix} \right]$$

For $v_2 = 1$

v_1 free

$$= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

The resulting matrix is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Explain informally what extra information this matrix records that ordinary eigenvectors miss.

The matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ encodes **transient, step-by-step dynamical coupling** (a Jordan chain), which pure eigenvectors completely miss because eigenvectors only detect static directions that map strictly into themselves.

The Information Missing from Ordinary Eigenvectors

- **Static Invariance vs. Directional Flow:**
  - An ordinary eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ satisfies $Tv_1 = 0 \cdot v_1 = 0$. It only identifies a static 1D subspace (the kernel) where vectors are crushed to zero immediately.
  - The second basis vector $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is a **generalized eigenvector**. It satisfies $(T - 0I)^2 v_2 = 0$, but $(T - 0I)v_2 \neq 0$.
  - Instead of mapping to a scalar multiple of itself, v_2 maps directly to v_1:

$$Tv_2 = v_1, \quad Tv_1 = 0$$

- **Nilpotent Shift Structure:**
  The off-diagonal 1 records a **chain of progression** (or shift operator):

$$v_2 \xrightarrow{T} v_1 \xrightarrow{T} 0$$

  Ordinary eigenvectors only tell you the "terminal state" ($v_1 \to 0$). The matrix entry 1 tells you that there is a 1-step "delay" or "queue" before reaching the kernel.
- **Dynamical and Differential Implications:**
  In continuous dynamical systems ($\dot{x} = Ax$), a diagonal matrix produces pure exponential decay or growth ($e^{\lambda t}$). A non-zero superdiagonal 1 in a Jordan block produces secular polynomial modulation:

Optional Exploratory Problem

16. What failure of diagonalization looks like. Let $T \in \mathcal{L}(\mathbb{C}^2)$ be defined by

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